

Assignment Details:

1. Prepare Terraform code using modules to deploy the following Azure resources for a production environment:
 - Azure Kubernetes Service (AKS)
 - Azure Container Registry (ACR)
 - Azure Key Vault
 - Azure App ServiceEnsure high availability, security, and networking best practices are followed.
2. Set up Azure DevOps CI/CD pipelines to deploy the above Terraform code.
3. Set up an Azure DevOps CI/CD pipeline to build a .NET application and deploy it to the Azure App Service.
4. Write Kubernetes YAML files for one microservice, including:
 - Deployment
 - Service
 - Ingress
 - Horizontal Pod Autoscaler (HPA)
 - StorageClass and PersistentVolumeClaim (PVC)
5. Prepare a Bill of Quantity (BOQ) for the above resources, considering monitoring and security costs for the UAE North region.

Note:

Use an Azure test subscription to demonstrate the deployment.

If an Azure test subscription is not available, prepare and share the YAML files (Terraform, Kubernetes, and Azure DevOps pipelines) for review.